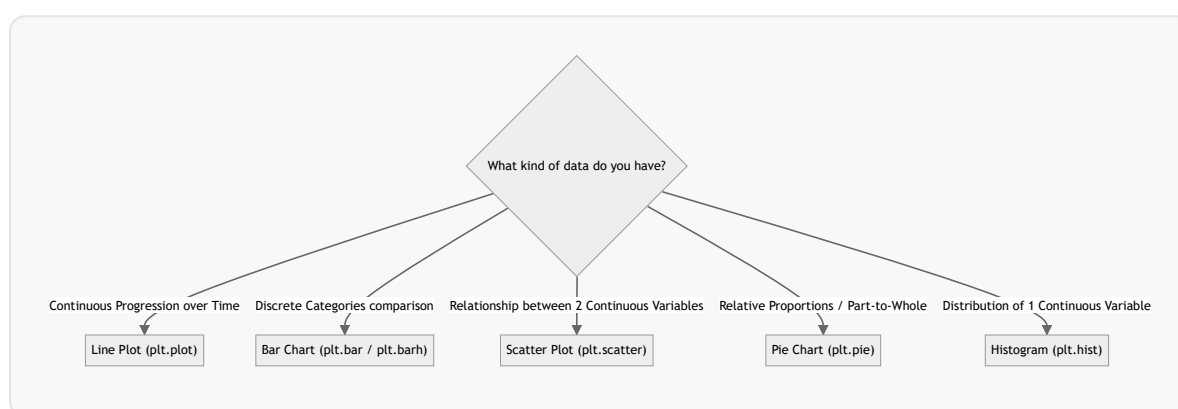


Day 18 - Lecture 18.8: Common Plots & Charts Taxonomy

1. Overview & Chart Taxonomy

Choosing the right chart is the single most important skill in data visualization. Using the wrong chart distorts findings and confuses stakeholders.



2. Summary Chart Selection Guide

Chart Type	Best Used When...	Avoid When...
Line Plot	Tracking trends and continuous sequential changes (time series)	Categories have no natural ordering
Bar Chart	Comparing numerical values across discrete categories	Data is continuous without groupings
Scatter Plot	Detecting correlations, clusters, and relationships between two continuous variables	Variables are pure categorical names
Pie Chart	Showing proportions of a whole for a very small number of categories (3–5 max)	More than 5 categories or comparing subtle differences
Histogram	Showing frequency distribution, skewness, and spread of continuous values	Data is discrete categories (use Bar Chart instead)

3. Key Takeaways

- **The Golden Rule:** Always identify whether your variables are **continuous (quantitative)** or **categorical (qualitative)** before selecting a chart type.